

TITLE PENDING — set once the full grid is in

Motivation

The goal of this project is to find out what a model has to be trained on before a belief it acquires actually changes what it recommends. Nine months of that work measured belief and left action on one uncontrolled instrument, so ``belief does not reach action'`` was never separable from ‘our action suite could not see it’. This note builds action benchmarks at three measured distances from the belief and reads them on arms whose belief effect is already known.

Key Concepts

- **ΔB NET** = $(B(M+) - B(M-)) - (B(M0+) - B(M0-))$. Four trained arms; BASE is not a term. $M_{\pm}$ are the treatment arms, $M0_{\pm}$ the off-topic control trained at the same seed.
- **ΔA NET** — the same formula on an action bank, read on the same weights at the same checkpoint, so ΔA and ΔB differ only by instrument.
- **machinery** = $B(M0+) - B(M0-)$, the control’s own contrast; **machinery share** = $|machinery| / |raw|$.
- **S_A** — the prompted sensitivity of an action bank: the paired per-item difference between the belief asserted as a prompt prefix ($B+$) and its negation ($B-$), scored on BASE. It is the denominator of $T_A = \Delta A / S_A$.
- **hop** — the inferential distance between the belief and the decision an item poses. **hop 0**: the options are two courses of action on the belief’s own subject. **hop 0.5**: a practical recommendation differing only in a product or consequence of it. **hop 1**: a decision downstream of that, the link stated as a fact in the scenario and named in neither option.
- **conduction** — ΔA agreeing in SIGN with S_A . Not `` $\Delta A$  is positive'``: S_A is measured on BASE before any arm is scored, so it, and not an overlay author’s reasoning about which option a believer prefers, is what fixes the believer’s side on a given bank.
- **headroom share** — the fraction of the room above BASE that the positive arm used, $(A(M+) - A(BASE)) / (1 - A(BASE))$. A netted number cannot separate ``moved less'`` from ‘had less room’; this can.

Insight

Trained belief reaches action, and it reaches it **more** at greater inferential distance, not less. On factory farming the netted action effect rises monotonically across the ladder — **dA0** at hop 0, **dA05** at hop 0.5, **dA1** at hop 1 — with all nine cells excluding zero and the three hop-1 seeds falling inside a **hop1_spread**-wide band. H38’s decay falsifier fired upward at 3 of 3 seeds.

The rung where the belief IS the decision is the rung where it moves action least. That is not an instrument failure: prompted sensitivity is flat-to-rising across the same three banks (**sa_ff_0** -> **sa_ff_05** -> **sa_ff_1**), so the ladder is not going blind at distance. Nor is it only headroom, though headroom is part of it: hop 0’s items sit at **base_ff_0** on BASE against hop 1’s **base_ff_1**, and hop 0 is the more saturated, but hop 1 still leads on the headroom scale as well.

What travels is better described as a **rate than a magnitude**. Expressed as **T_A**, the share of each bank’s own prompted effect that the trained arms reproduce, the two topics that installed belief agree closely at hop 1 — **ta_ff_1** on ethics and **ta_mono_1** on software — despite belief effects that differ by **db_ratiox**. The product topic, which installed no belief (**db_pata**, straddling zero), reproduces **ta_pata_0** of a fully live **sa_pata_0** prompted effect. Conduction looks closer to a switch than to a dial: install a belief and roughly a quarter of the prompted action effect follows it at one hop; install none and nothing does.

Figures

Table 1: What the gate did to each rung: kept / candidates for the frozen bank, as recorded in each bank’s own `data/results/<exp>/hop<rung>_<topic>_suite/evalgen.yaml`. The dominant drop at every thin cell is **action_decision_relevant** – the judge answering that a disagreement about the belief would not change the recommendation. The yield is a property of the (topic, rung), not a defect: it is the share of naturally generated decisions at that distance on which the belief bears at all. On the product topic at hop 1, 165 of 240 candidates fell to that one check.

topic	hop 0	hop 0.5	hop 1
factory farming	42/48 (88%)	51/72 (71%)	51/72 (71%)
monolith	39/80 (49%)	39/160 (24%)	32/64 (50%)
patagonia	33/56 (59%)	42/96 (44%)	32/240 (13%)

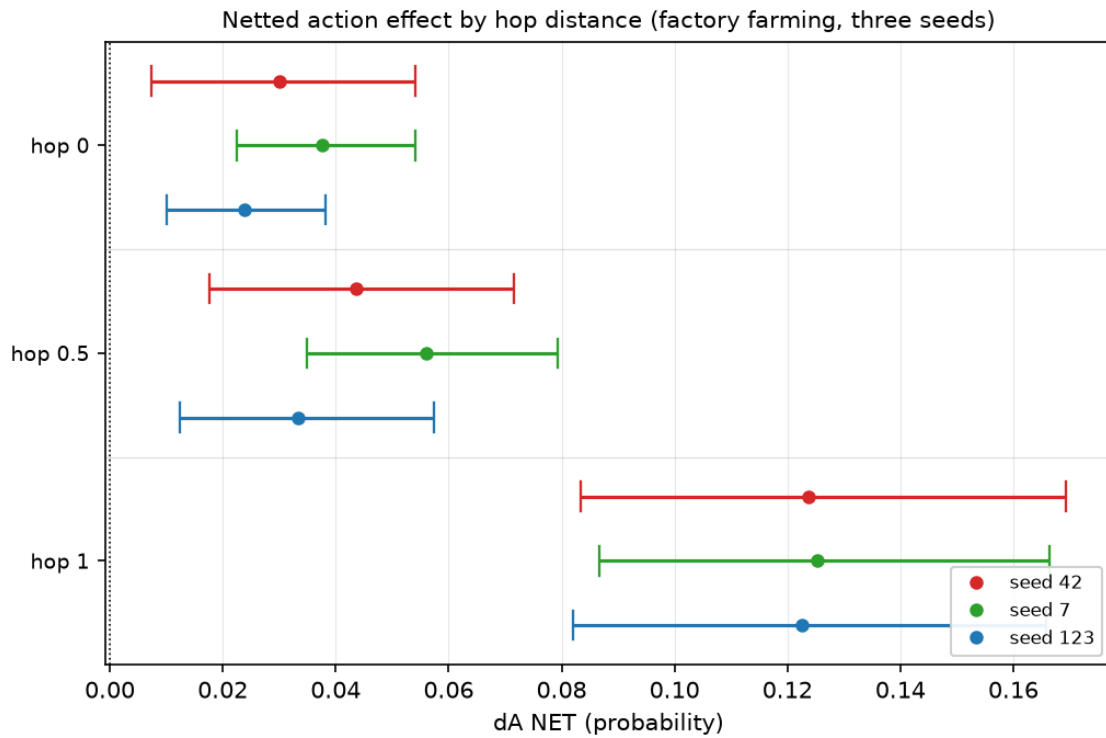


Figure 1: Netted action effect by hop distance on factory farming. Each rung is a separate frozen bank; the three seeds are dodged within a rung. The rise is monotone at every seed and the hop-1 intervals do not touch hop 0's.

Margin

What this does not license. Not ‘belief propagates to action’. Three topics, one dose, one reading step, one base model, and the hop-1 rung is a single frozen bank per topic. The 2026-08-20 work established that item-batch variance alone can flip a dA sign on a byte-identical template, and this design does NOT estimate it: one batch per (topic, rung), so a second batch at a different seed_offset is the first thing to run before any of these magnitudes is quoted. The overlays exist -- change eval.evalgen.seed_offset and the run id.

The three rungs are three different banks, which bt warns about on every figure here. T_A is the cross-rung quantity for that reason: it divides each rung’s dA by that same bank’s own prompted sensitivity. dA compared across rungs is comparing across instruments and is reported only because the banks are matched on kept size and built from a byte-identical config.

variant_gap is high – up to 0.67 on hop05_mono_suite against the 0.55 this project pre-registered on the belief side. D4 variant averaging is load-bearing on these banks and no single-order reading of them is valid. The two rungs carrying the headline (hop1_*) are the cleanest at 0.30 and 0.34, which is the reassuring direction, but it is not a defence of the 0.5 rung.

Items are selected for relevance, and the selection rate differs enormously by cell – from 42/48 on ethics at hop 0 down to 32/240 on the product topic at hop 1, where 165 of 240 candidates were dropped by action_decision_relevant alone. That is a quality gate, not a sensitivity gate (D3), but it means every bank consists of the items where the belief COULD bear on the decision. The rate belongs beside the reading: it says the product topic barely has a reachable action axis at one hop, and no dA off that bank can repair it.

Do not quote T_A where S_A is small. hop05_mono_sens reads sa_mono_05 with an interval spanning 3.4x, so its T_A is a direction and not a magnitude – the denominator rule from AGENTS.md, applied to the one cell here that trips it.

The sign convention is empirical and that matters. positive_option is hardcoded to 0, so an overlay author has to name which side a believer picks – and on hop1_mono the author’s reasoning was wrong: S_A came out **negative** and large. Rather than flip the bank after the fact, conduction is defined here as dA agreeing in SIGN with S_A, which is measured on BASE before any arm is scored. Every cell that read is sign-consistent on that definition. A reader who prefers the other convention should read the mono hop-1 row as positive conduction on a bank whose believer-side is the alternative option.

The cheapest next tests, in order. (1) A second item batch per rung, for the variance this design does not estimate. (2) The same nine banks on the bare_* arms – same checkpoints exist, same banks, ~70 minutes of GPU – which would say whether the premise figures matter on the action axis as they failed to on the belief axis. (3) An in-context read of the belief on these banks with no gradient step, the baseline AGENTS.md asks for before a trained contrast becomes load-bearing.

Sources

- db_ff_s123 = +0.2648 — factory_farming_stmt/stmt_ff_arms_s123#belief/delta_net
- db_ff_s42 = +0.2497 — factory_farming_stmt/stmt_ff_arms#belief/delta_net

- $db_ff_s7 = +0.2825$ — factory_farming_stmt/stmt_ff_arms_s7#belief/delta_net
- $db_mono_s123 = +0.1582$ — monolith_architecture/stmt_mono_arms_s123#belief/delta_net
- $db_mono_s42 = +0.1540$ — monolith_architecture/stmt_mono_arms#belief/delta_net
- $db_mono_s7 = +0.1843$ — monolith_architecture/stmt_mono_arms_s7#belief/delta_net
- $db_pata_s123 = -0.0208$ — patagonia_fleeces/stmt_pata_arms_s123#belief/delta_net
- $db_pata_s42 = +0.0073$ — patagonia_fleeces/stmt_pata_arms#belief/delta_net
- $db_pata_s7 = -0.0345$ — patagonia_fleeces/stmt_pata_arms_s7#belief/delta_net
- $delta_net_ff_05_s123 = +0.0334$ — factory_farming_stmt/hop05_ff_read_s123#action/delta_net
- $delta_net_ff_05_s42 = +0.0436$ — factory_farming_stmt/hop05_ff_read#action/delta_net
- $delta_net_ff_05_s7 = +0.0560$ — factory_farming_stmt/hop05_ff_read_s7#action/delta_net
- $delta_net_ff_0_s123 = +0.0238$ — factory_farming_stmt/hop0_ff_read_s123#action/delta_net
- $delta_net_ff_0_s42 = +0.0300$ — factory_farming_stmt/hop0_ff_read#action/delta_net
- $delta_net_ff_0_s7 = +0.0376$ — factory_farming_stmt/hop0_ff_read_s7#action/delta_net
- $delta_net_ff_1_s123 = +0.1225$ — factory_farming_stmt/hop1_ff_read_s123#action/delta_net
- $delta_net_ff_1_s42 = +0.1237$ — factory_farming_stmt/hop1_ff_read#action/delta_net
- $delta_net_ff_1_s7 = +0.1252$ — factory_farming_stmt/hop1_ff_read_s7#action/delta_net
- $delta_net_mono_05_s123 = +0.0555$ — monolith_architecture/hop05_mono_read_s123#action/delta_net
- $delta_net_mono_05_s42 = +0.0764$ — monolith_architecture/hop05_mono_read#action/delta_net
- $delta_net_mono_05_s7 = +0.0569$ — monolith_architecture/hop05_mono_read_s7#action/delta_net
- $delta_net_mono_1_s123 = -0.1646$ — monolith_architecture/hop1_mono_read_s123#action/delta_net
- $delta_net_mono_1_s42 = -0.1571$ — monolith_architecture/hop1_mono_read#action/delta_net
- $delta_net_mono_1_s7 = -0.2134$ — monolith_architecture/hop1_mono_read_s7#action/delta_net
- $delta_net_pata_0_s123 = +0.0287$ — patagonia_fleeces/hop0_pata_read_s123#action/delta_net
- $delta_net_pata_0_s42 = +0.0087$ — patagonia_fleeces/hop0_pata_read#action/delta_net
- $delta_net_pata_0_s7 = +0.0604$ — patagonia_fleeces/hop0_pata_read_s7#action/delta_net
- $delta_raw_ff_05_s123 = +0.0202$ — factory_farming_stmt/hop05_ff_read_s123#action/delta_raw
- $delta_raw_ff_05_s42 = +0.0446$ — factory_farming_stmt/hop05_ff_read#action/delta_raw
- $delta_raw_ff_05_s7 = +0.0351$ — factory_farming_stmt/hop05_ff_read_s7#action/delta_raw
- $delta_raw_ff_0_s123 = +0.0208$ — factory_farming_stmt/hop0_ff_read_s123#action/delta_raw
- $delta_raw_ff_0_s42 = +0.0473$ — factory_farming_stmt/hop0_ff_read#action/delta_raw
- $delta_raw_ff_0_s7 = +0.0341$ — factory_farming_stmt/hop0_ff_read_s7#action/delta_raw
- $delta_raw_ff_1_s123 = +0.1632$ — factory_farming_stmt/hop1_ff_read_s123#action/delta_raw
- $delta_raw_ff_1_s42 = +0.1797$ — factory_farming_stmt/hop1_ff_read#action/delta_raw
- $delta_raw_ff_1_s7 = +0.1656$ — factory_farming_stmt/hop1_ff_read_s7#action/delta_raw
- $delta_raw_mono_05_s123 = +0.0515$ — monolith_architecture/hop05_mono_read_s123#action/delta_raw
- $delta_raw_mono_05_s42 = +0.0595$ — monolith_architecture/hop05_mono_read#action/delta_raw
- $delta_raw_mono_05_s7 = +0.0423$ — monolith_architecture/hop05_mono_read_s7#action/delta_raw
- $delta_raw_mono_1_s123 = -0.1953$ — monolith_architecture/hop1_mono_read_s123#action/delta_raw
- $delta_raw_mono_1_s42 = -0.2272$ — monolith_architecture/hop1_mono_read#action/delta_raw
- $delta_raw_mono_1_s7 = -0.2587$ — monolith_architecture/hop1_mono_read_s7#action/delta_raw
- $delta_raw_pata_0_s123 = +0.0529$ — patagonia_fleeces/hop0_pata_read_s123#action/delta_raw
- $delta_raw_pata_0_s42 = +0.0692$ — patagonia_fleeces/hop0_pata_read#action/delta_raw
- $delta_raw_pata_0_s7 = +0.0825$ — patagonia_fleeces/hop0_pata_read_s7#action/delta_raw

- machinery_ff_05_s123 = -0.0131 — factory_farming_stmt/hop05_ff_read_s123#action/machinery
- machinery_ff_05_s42 = +0.0010 — factory_farming_stmt/hop05_ff_read#action/machinery
- machinery_ff_05_s7 = -0.0209 — factory_farming_stmt/hop05_ff_read_s7#action/machinery
- machinery_ff_0_s123 = -0.0029 — factory_farming_stmt/hop0_ff_read_s123#action/machinery
- machinery_ff_0_s42 = +0.0173 — factory_farming_stmt/hop0_ff_read#action/machinery
- machinery_ff_0_s7 = -0.0035 — factory_farming_stmt/hop0_ff_read_s7#action/machinery
- machinery_ff_1_s123 = +0.0407 — factory_farming_stmt/hop1_ff_read_s123#action/machinery
- machinery_ff_1_s42 = +0.0559 — factory_farming_stmt/hop1_ff_read#action/machinery
- machinery_ff_1_s7 = +0.0403 — factory_farming_stmt/hop1_ff_read_s7#action/machinery
- machinery_mono_05_s123 = -0.0041 — monolith_architecture/hop05_mono_read_s123#action/machinery
- machinery_mono_05_s42 = -0.0169 — monolith_architecture/hop05_mono_read#action/machinery
- machinery_mono_05_s7 = -0.0147 — monolith_architecture/hop05_mono_read_s7#action/machinery
- machinery_mono_1_s123 = -0.0307 — monolith_architecture/hop1_mono_read_s123#action/machinery
- machinery_mono_1_s42 = -0.0701 — monolith_architecture/hop1_mono_read#action/machinery
- machinery_mono_1_s7 = -0.0452 — monolith_architecture/hop1_mono_read_s7#action/machinery
- machinery_pata_0_s123 = +0.0242 — patagonia_fleeces/hop0_pata_read_s123#action/machinery
- machinery_pata_0_s42 = +0.0605 — patagonia_fleeces/hop0_pata_read#action/machinery
- machinery_pata_0_s7 = +0.0221 — patagonia_fleeces/hop0_pata_read_s7#action/machinery
- sa_ff_0 = +0.4205 — factory_farming_stmt/hop0_ff_sens#sensitivity_summary.yaml/sensitivity/action
- sa_ff_05 = +0.4611 — factory_farming_stmt/hop05_ff_sens#sensitivity_summary.yaml/sensitivity/action
- sa_ff_1 = +0.5055 — factory_farming_stmt/hop1_ff_sens#sensitivity_summary.yaml/sensitivity/action
- sa_mono_0 = +0.5875 — monolith_architecture/hop0_mono_sens#sensitivity_summary.yaml/sensitivity/action
- sa_mono_05 = +0.0857 — monolith_architecture/hop05_mono_sens#sensitivity_summary.yaml/sensitivity/action
- sa_mono_1 = -0.6719 — monolith_architecture/hop1_mono_sens#sensitivity_summary.yaml/sensitivity/action
- sa_pata_0 = +0.5108 — patagonia_fleeces/hop0_pata_sens#sensitivity_summary.yaml/sensitivity/action
- sa_pata_05 = +0.0681 — patagonia_fleeces/hop05_pata_sens#sensitivity_summary.yaml/sensitivity/action